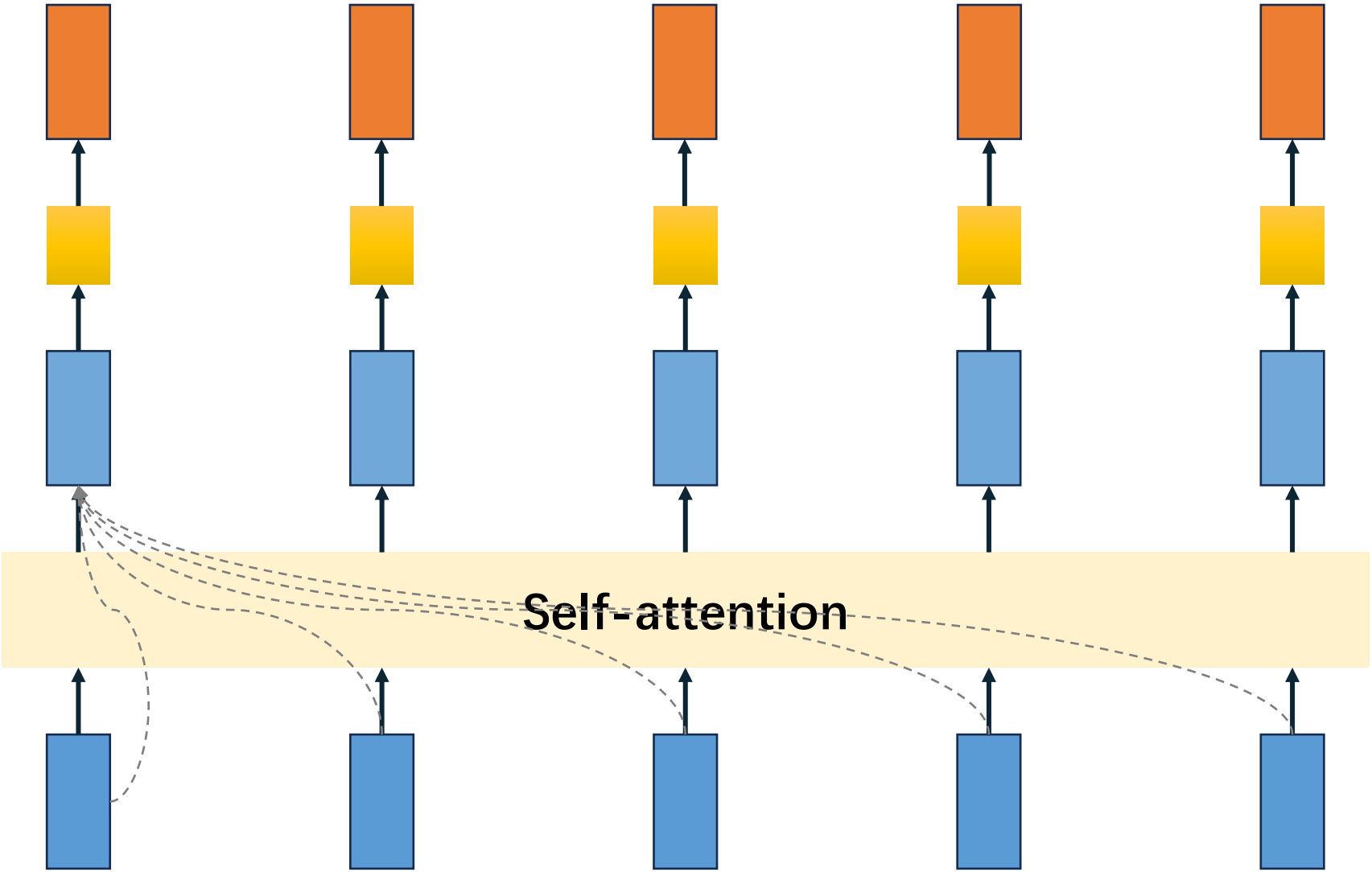
第7.4节 Transformer

Structure of Transformer

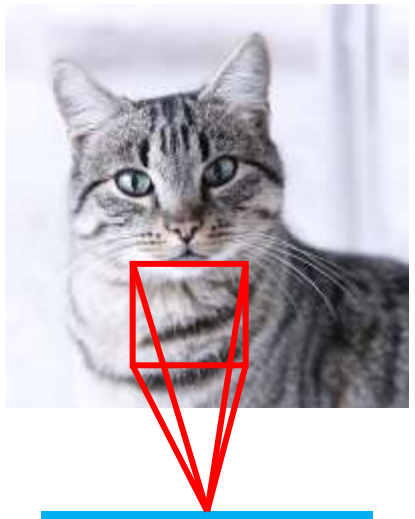
前情回顾： Self-attention

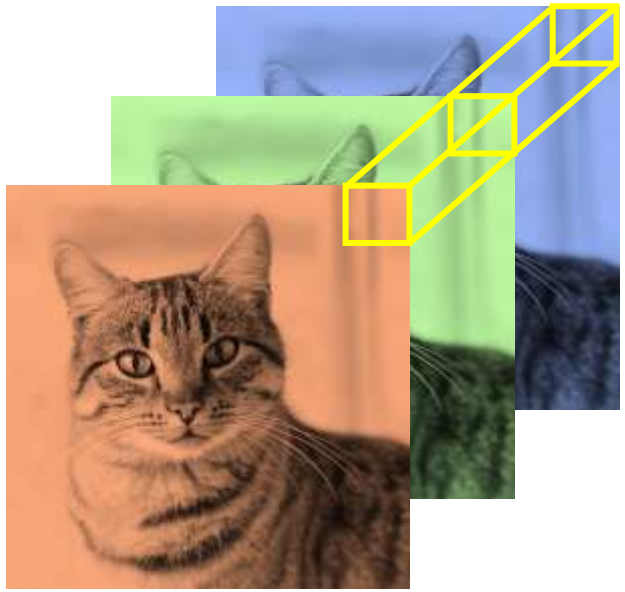
考虑整个
上下文内容

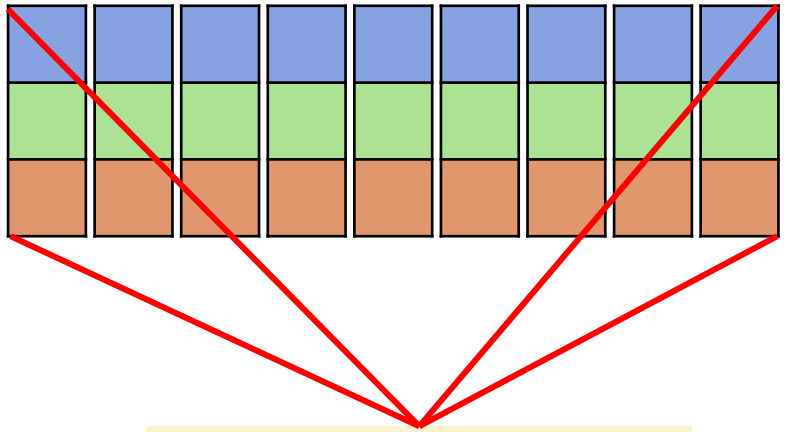


前情回顾：为什么说“Attention is all you need？”

Self-attention vs CNN

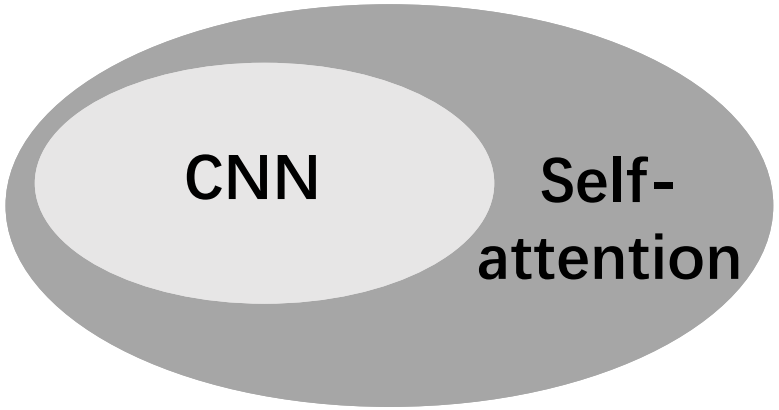






CNN

关注局部特征

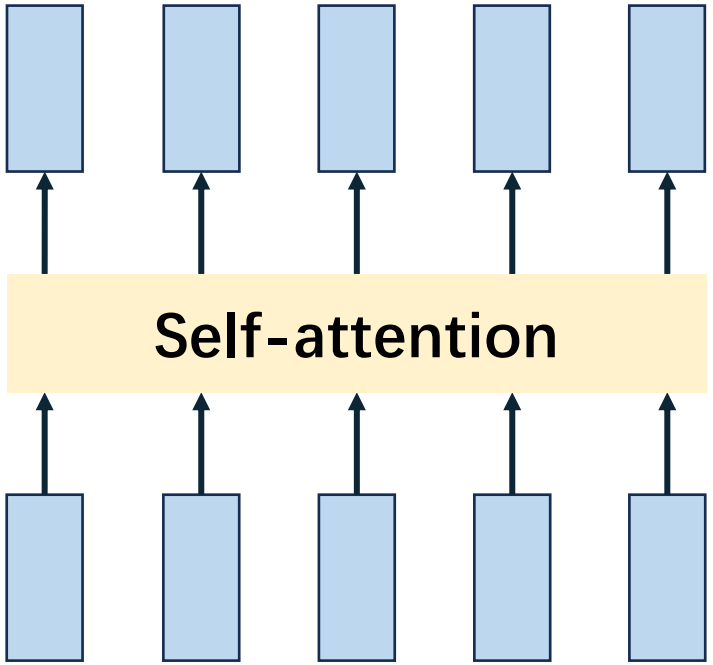


Self-attention

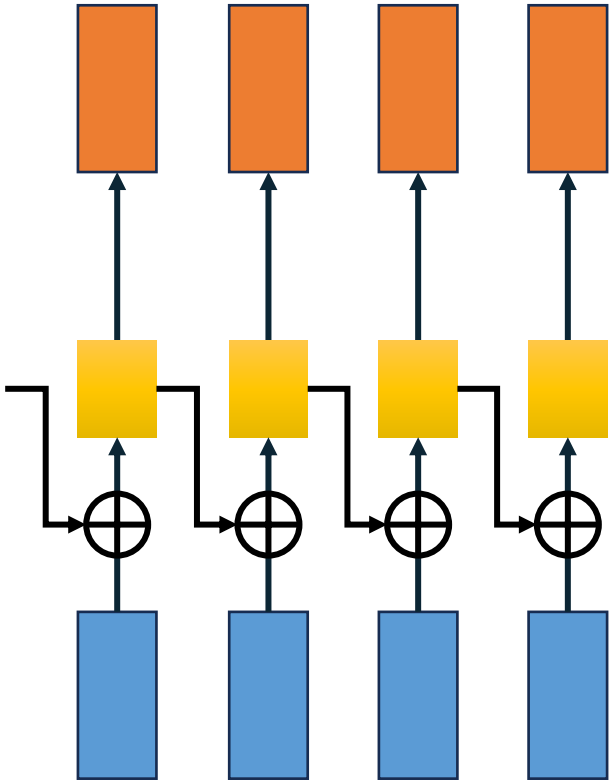
纵观全局特征
关注局部特征

前情回顾：为什么说“Attention is all you need？”

Self-attention vs RNN



可以并行执行



只能顺序执行

Transformer

Attention Is All You Need

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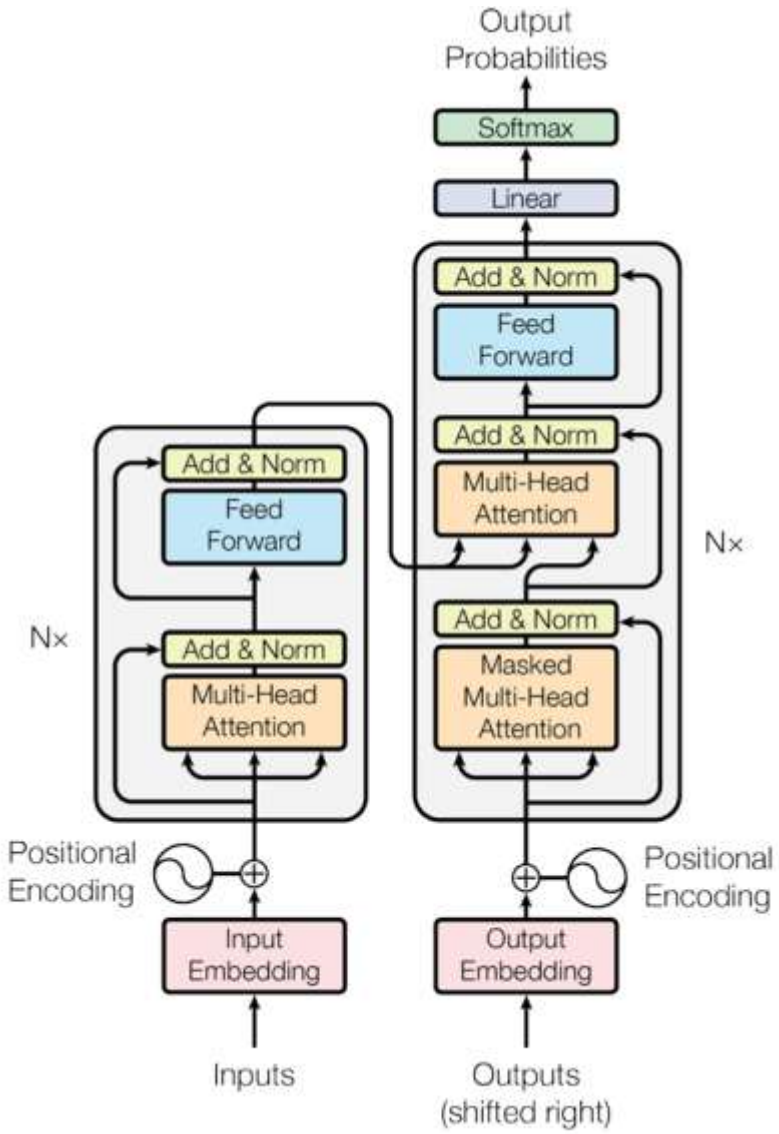


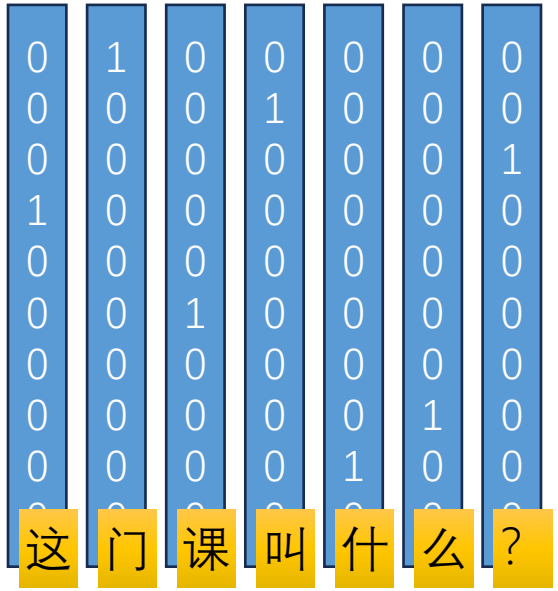
序列到序列问题

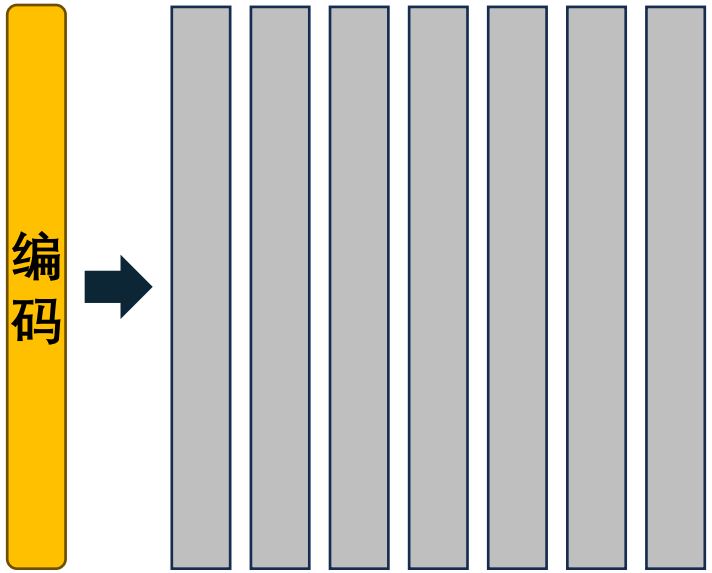
这门课叫什么？

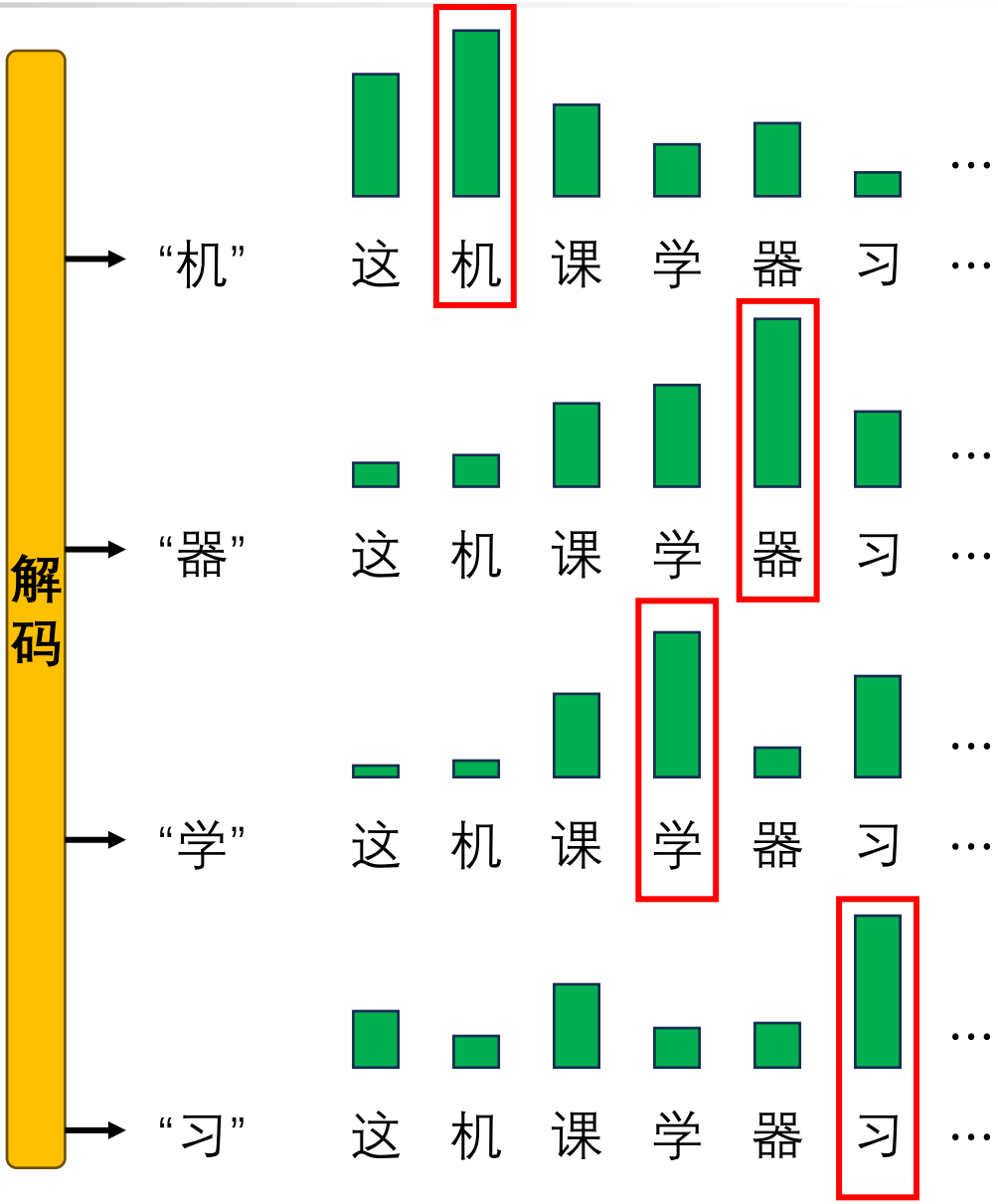
机器学习

“这门课叫什么？”

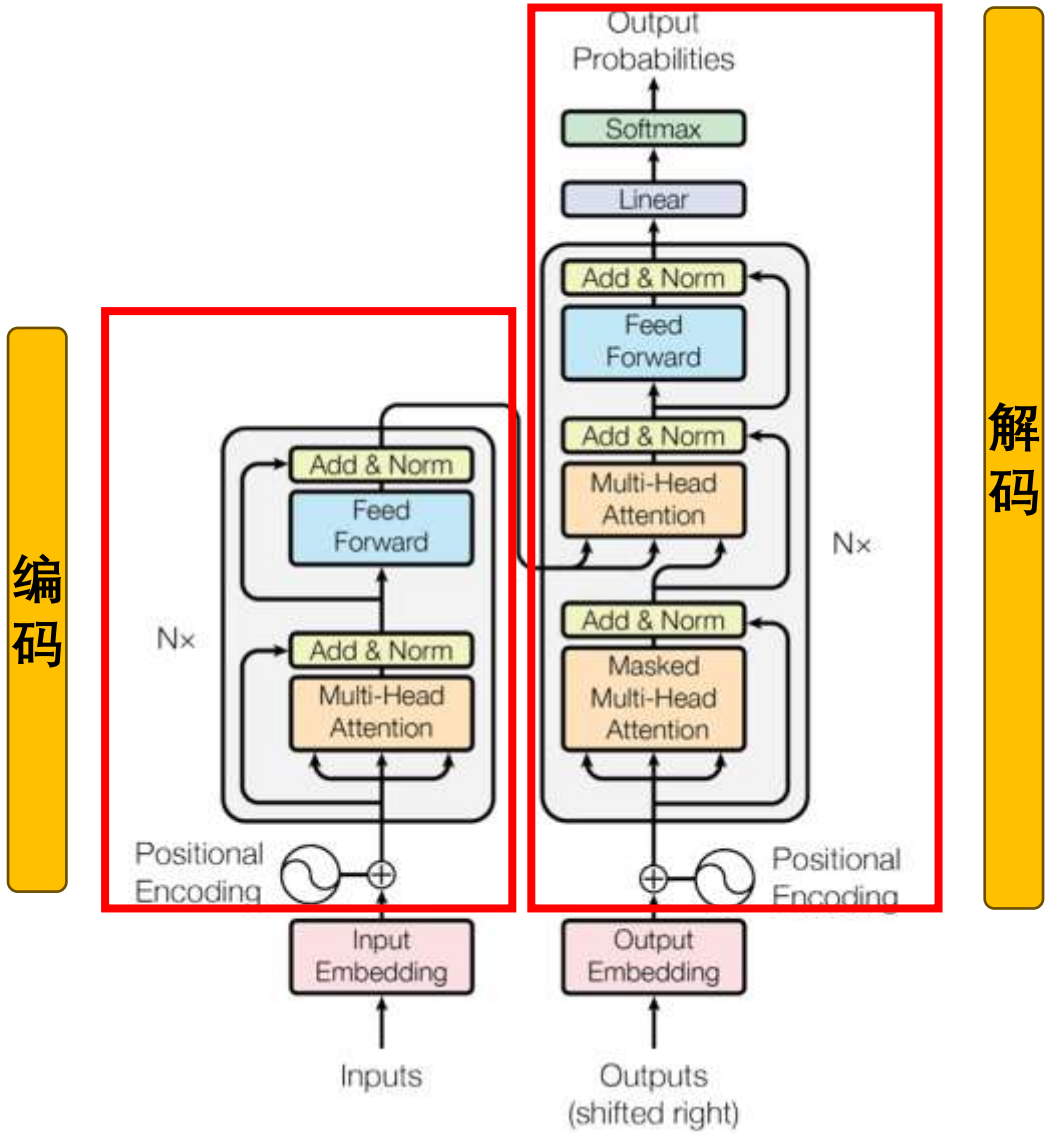
向量表示（词向量）



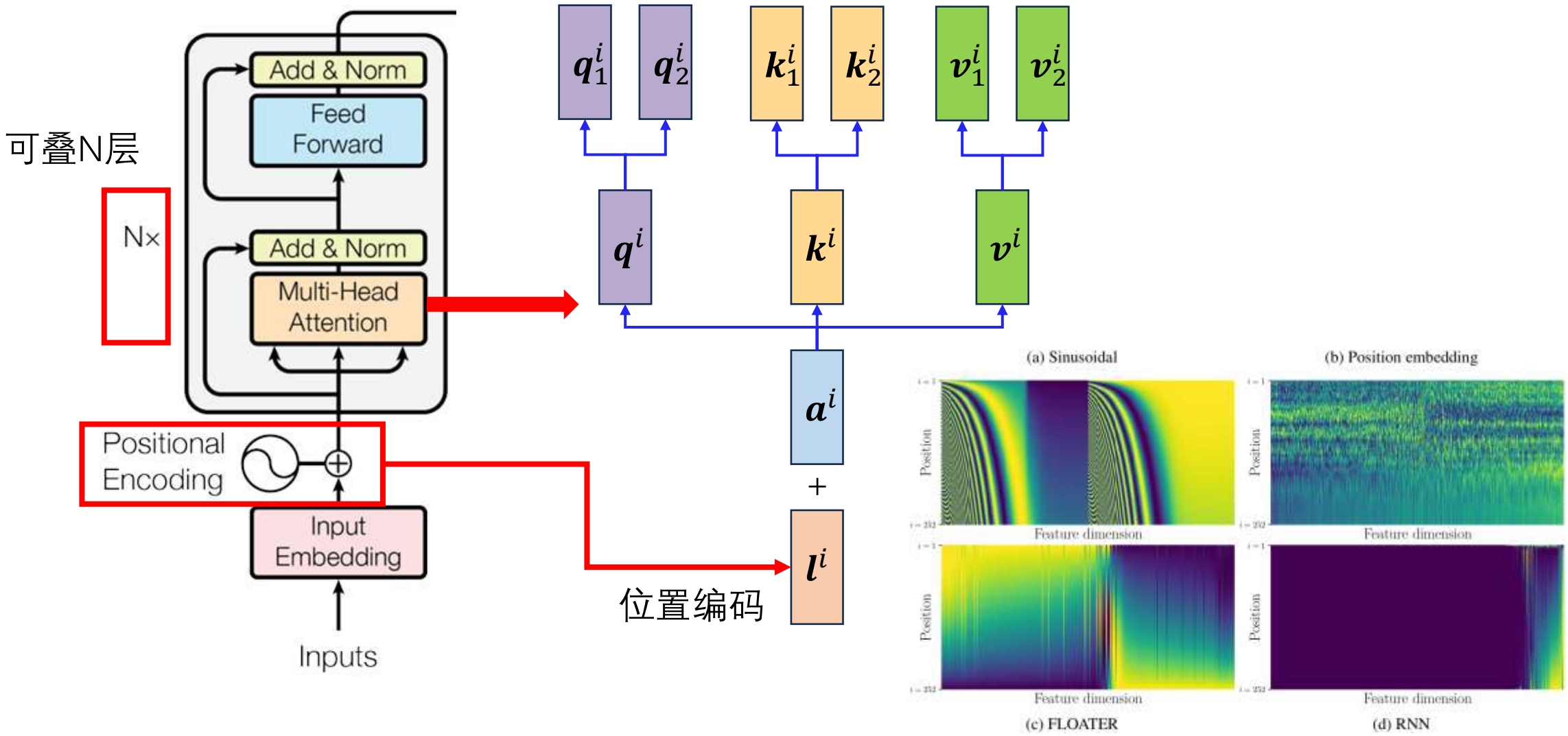




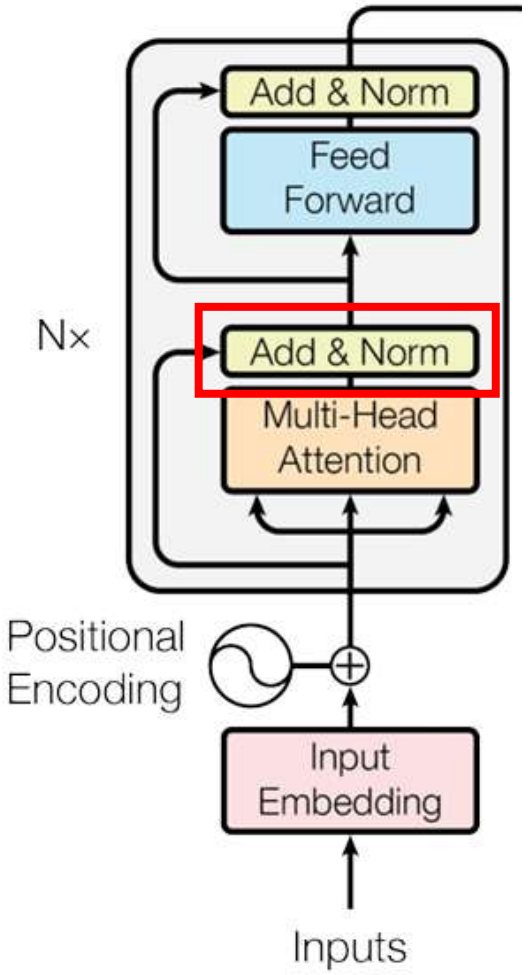
Transformer



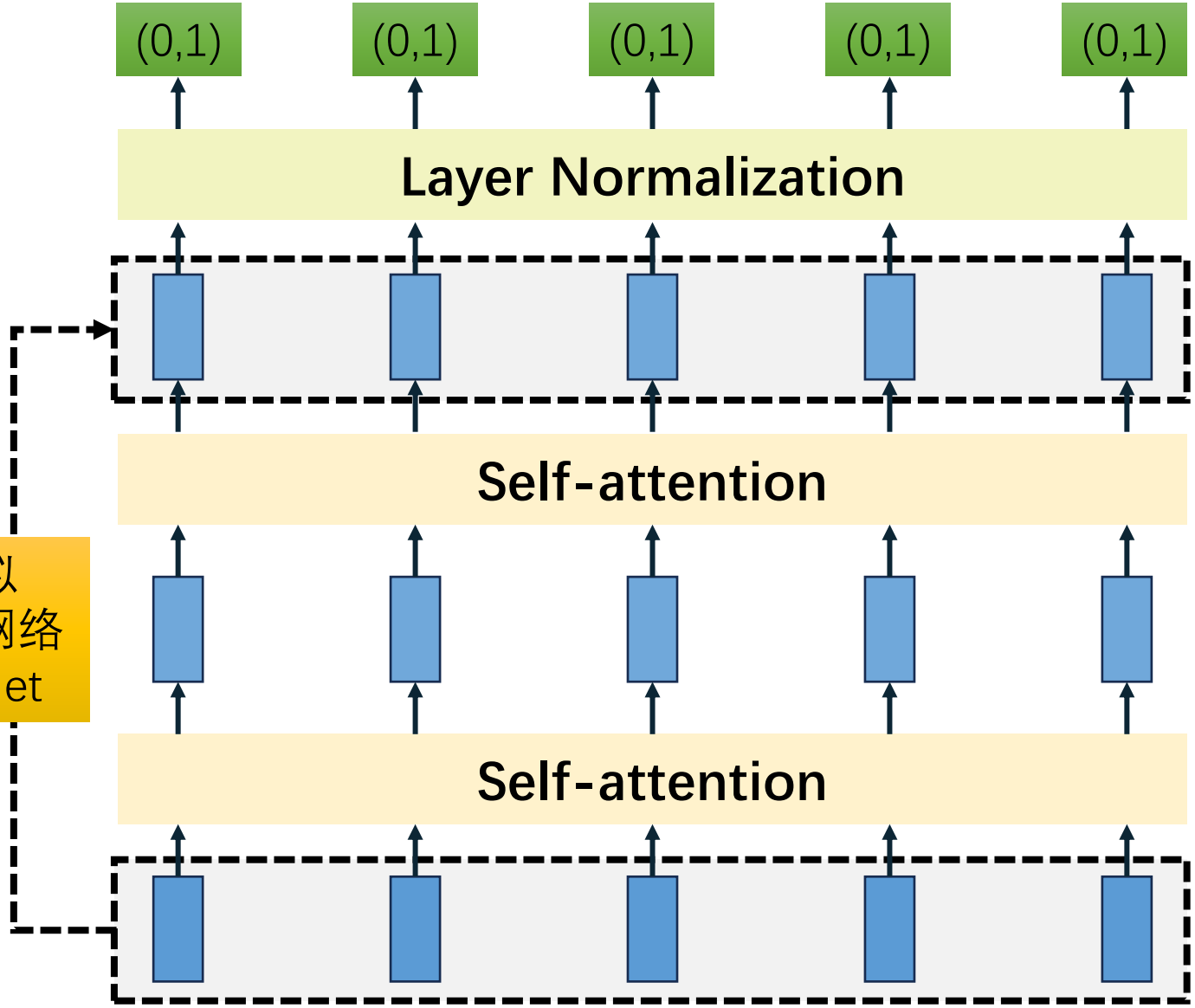
Transformer: Encoder



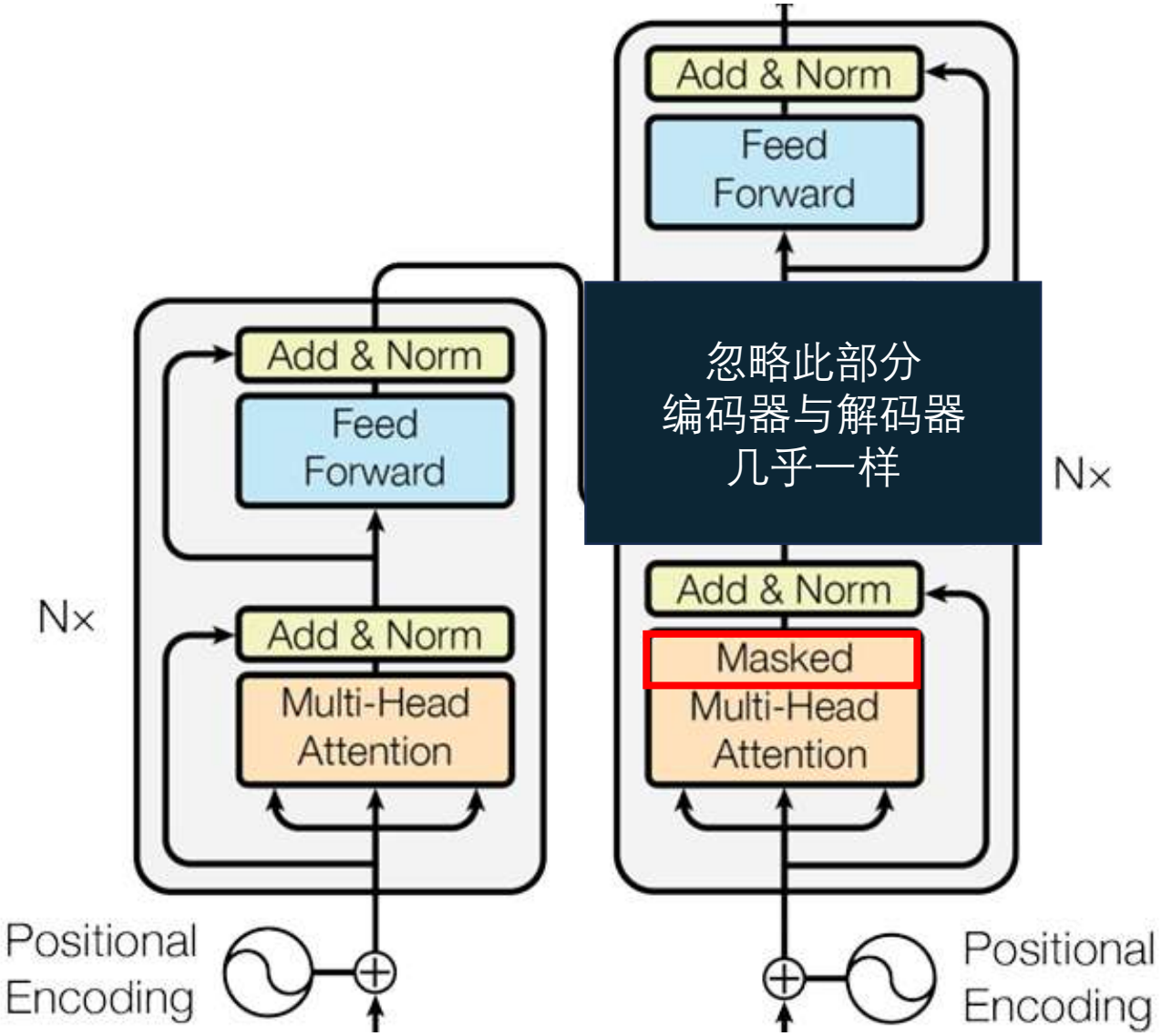
Transformer: Encoder



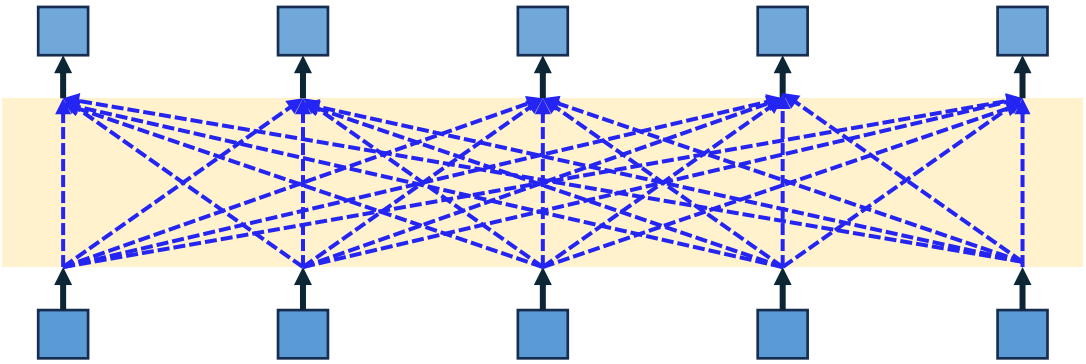
类似
残差网络
ResNet



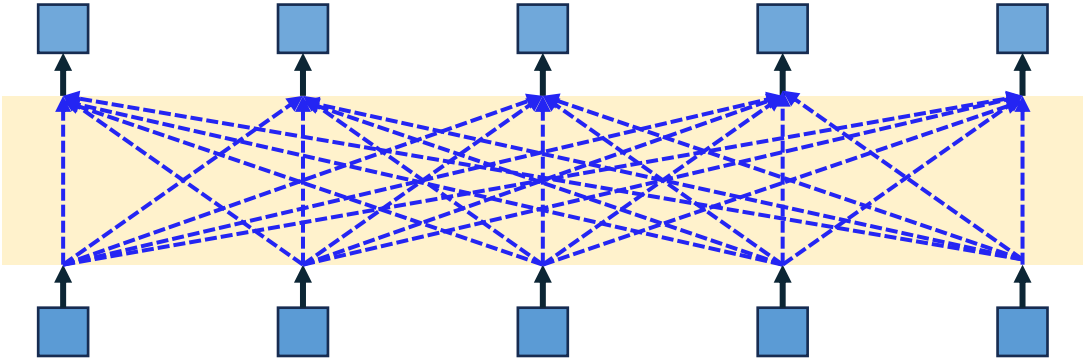
Transformer: Decoder



忽略此部分
编码器与解码器
几乎一样



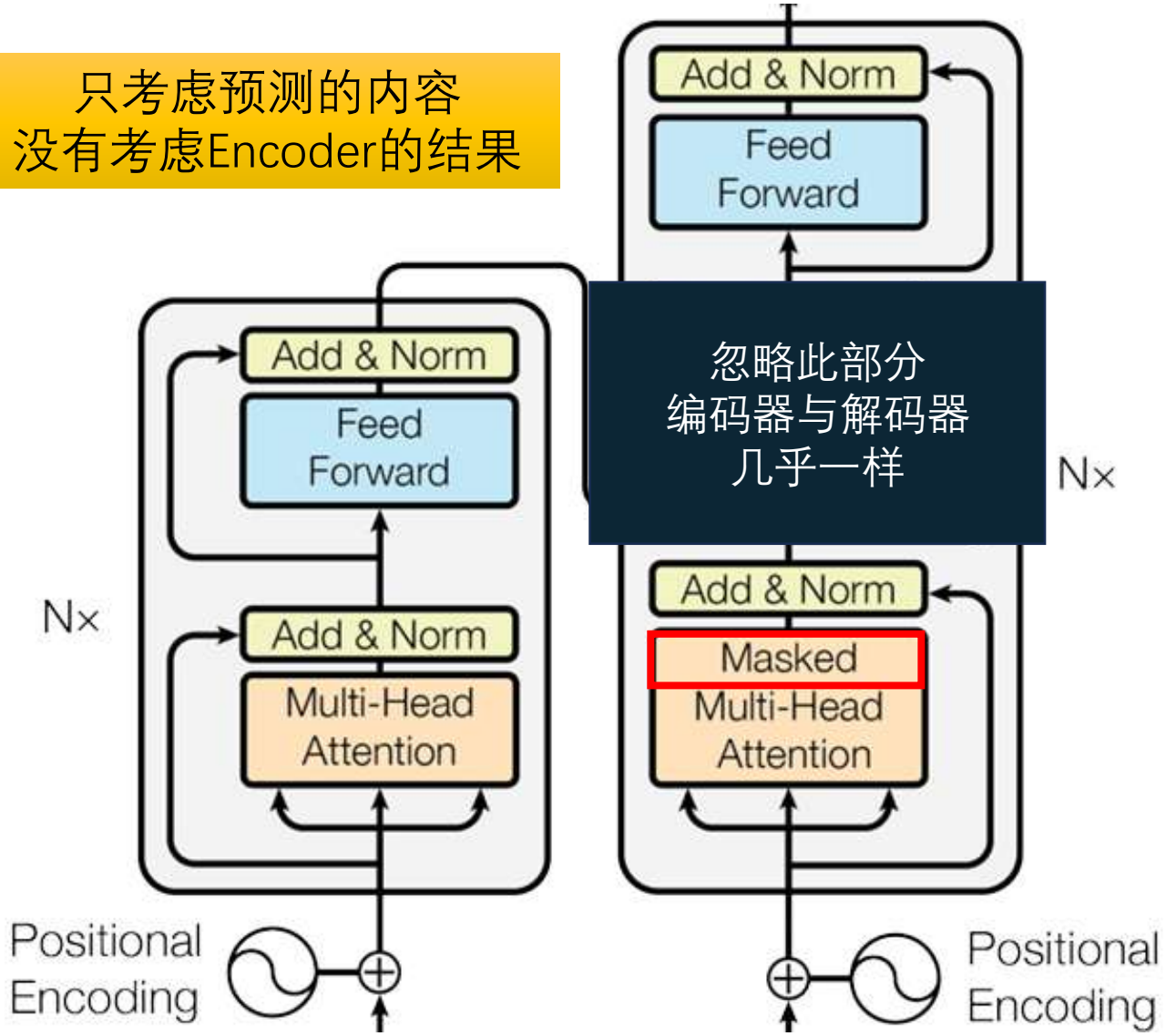
Self-attention



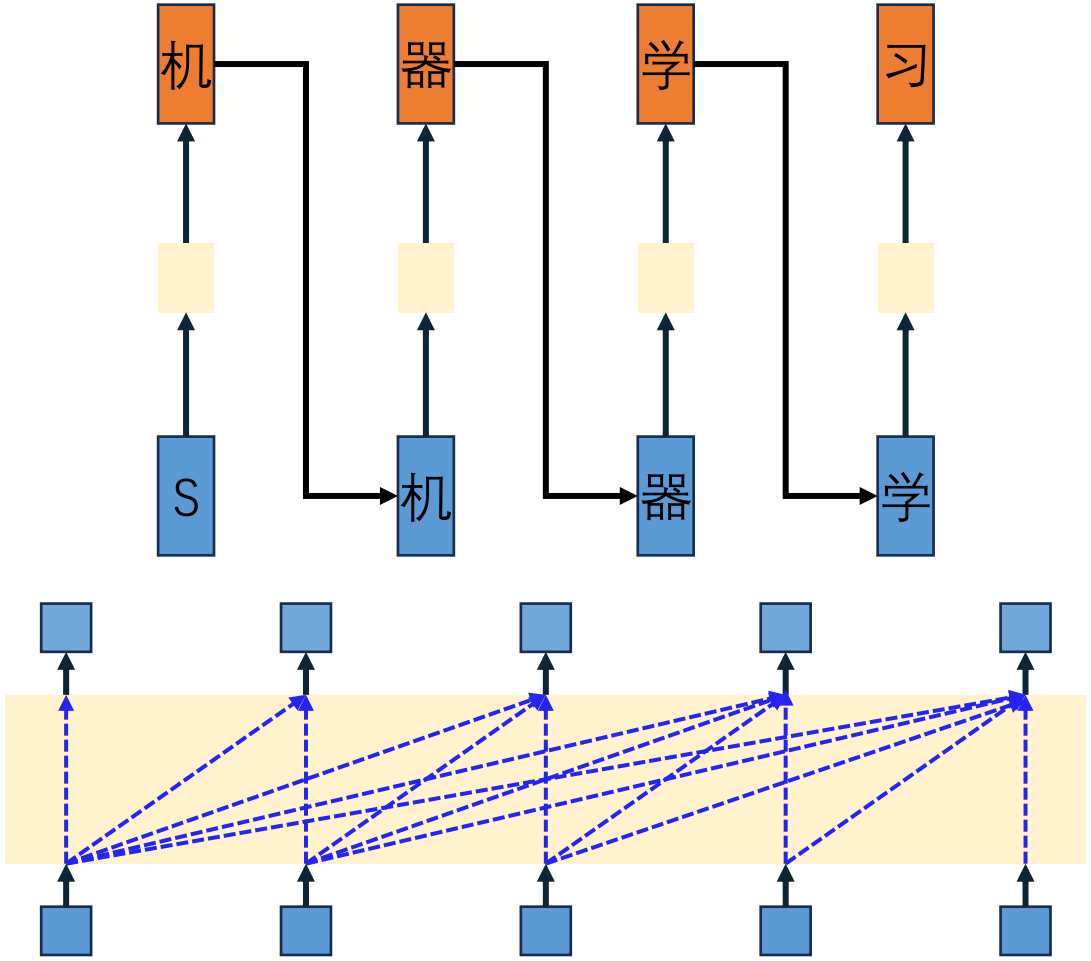
Masked Self-attention

Transformer: Decoder

只考虑预测的内容
没有考虑Encoder的结果



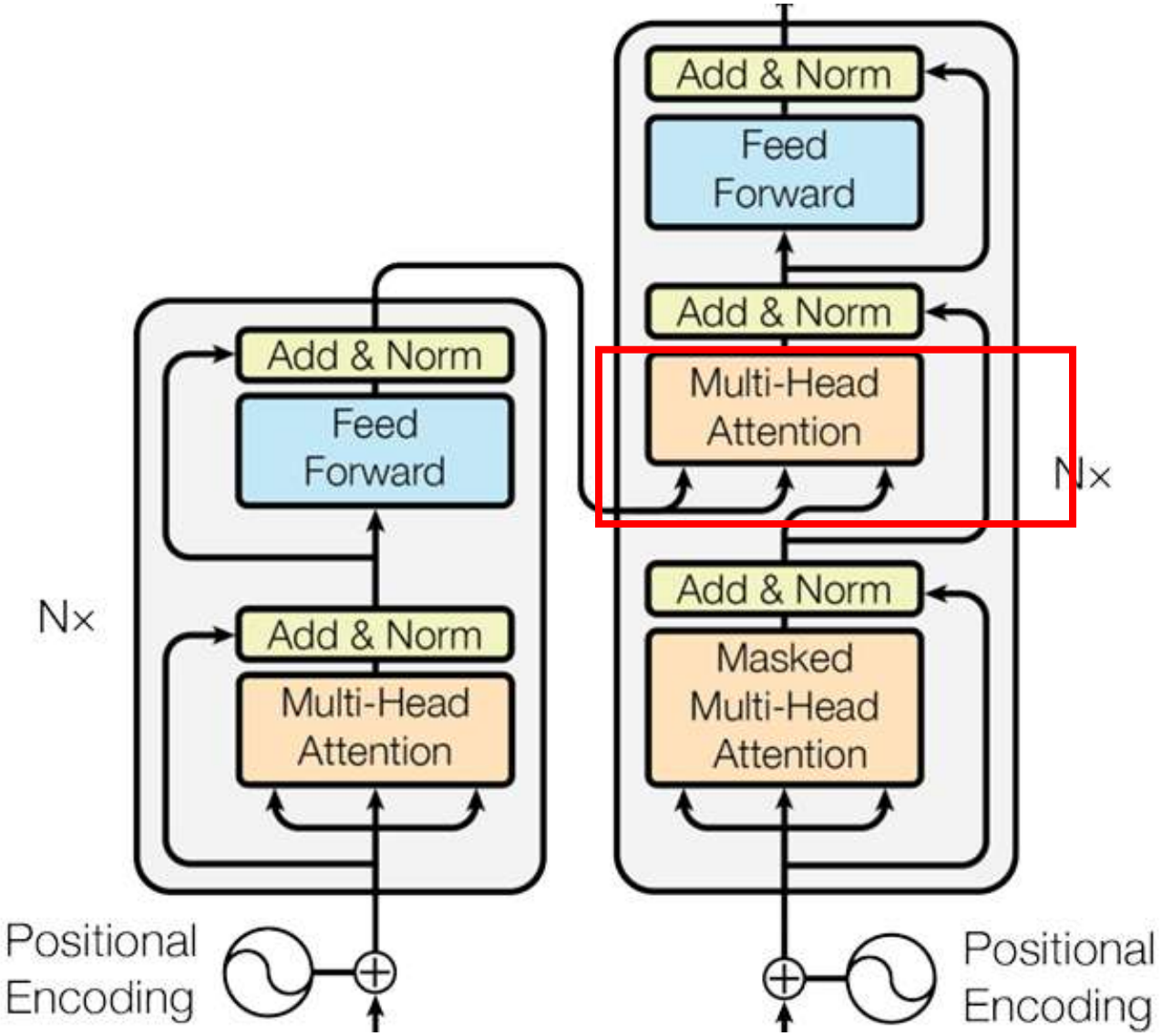
忽略此部分
编码器与解码器
几乎一样



Masked Self-attention

限制当前输出只能参考之前时刻的内容

Transformer: Decoder

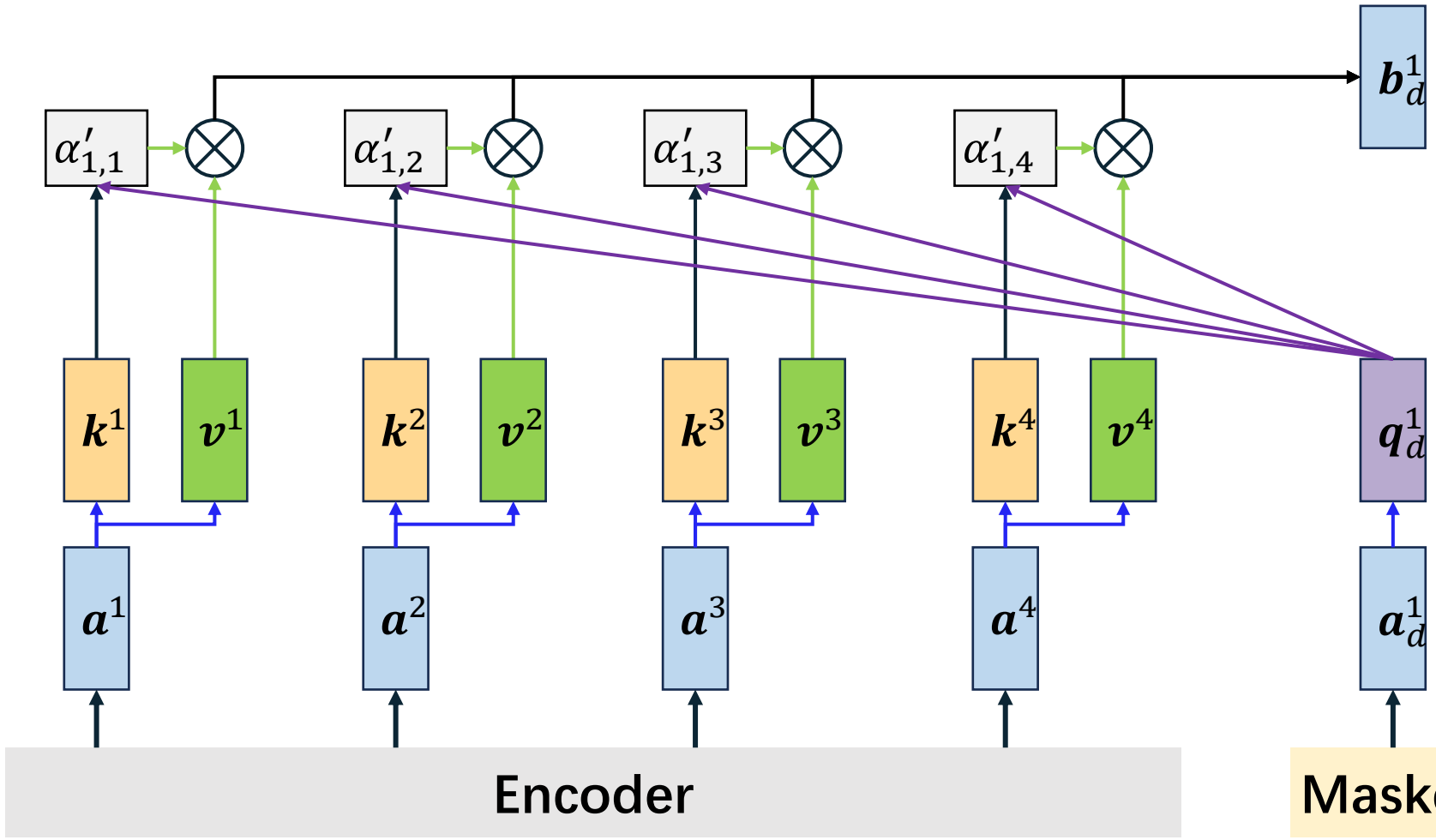


Cross Attention

Transformer: Decoder

Cross Attention

计算Decoder的问题（Query）
和Encoder的答案（Key）之间的相关性

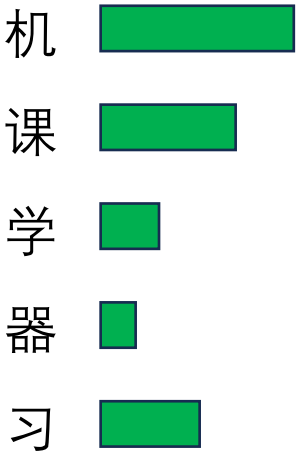


生成式大型语言模型（LLM）在做什么？

文字接龙

这门课叫什么？

机器学习



“这门课叫什么？”	LLM	机
“这门课叫什么？机”	LLM	器
“这门课叫什么？机器”	LLM	学
“这门课叫什么？机器学”	LLM	习
“这门课叫什么？机器学习”	LLM	END

生成式大型语言模型（LLM）在做什么？

文字接龙

句子 → LLM → 字 字 字 字 字

图片 → LLM → 像素 像素 像素 像素 像素
像素 像素 像素 像素 像素
像素 像素 像素 像素 像素
像素 像素 像素 像素 像素

相比传统的机器学习模型LLM是一个全才

LLM怎么就这么强？

Scaling Laws for Neural Language Models

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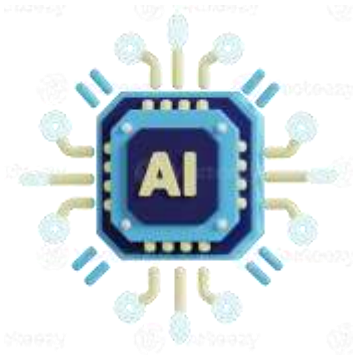
https://arxiv.org/pdf/2001.08361

LLM的性能随着模型规模和数据量增加而提高

- 出现**知识涌现**
- 边际效益递减
- 训练成本增高

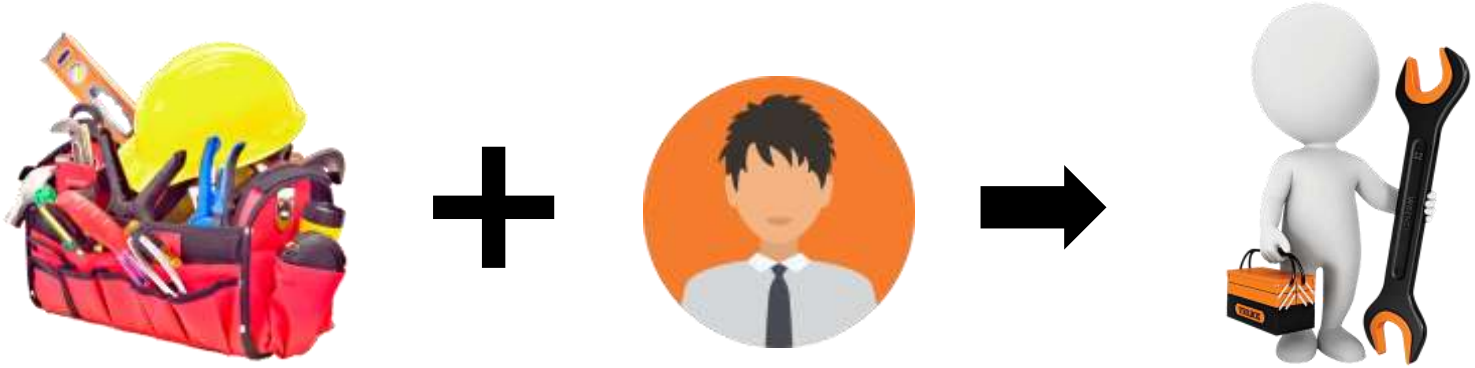
Improving Performance of LLM

ChatGPT、Gemini、
Kimi、Claude



在使用LLM时还能如何提升能力呢?

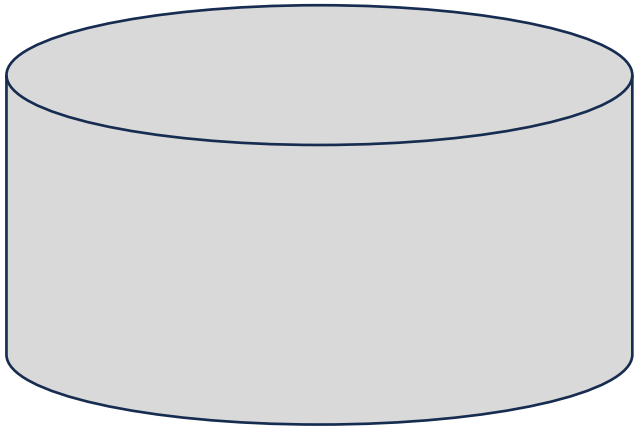
SVM、Decision Tree
FNN、CNN、RNN



Improving Performance of LLM

你可以……　　训练一个巨大的LLM

XXXFormer



准备一大堆数据

准备一大堆GPU

需要极高的成本

Improving Performance of LLM

还有一些超越代际的技术

思维链
Chain of Thought
CoT

工作流
Workflow

Level UP

LLM

智能体
Agent

模型微调
Fine-tune

Chain of Thought

Let think step by step

让AI把推理结果一步一步写出来

屋子里有一些鸡和兔子，总共有23个头，67只腿。问：有多少只鸡和多少只兔子？直接告诉我答案

鸡有13只，兔子有10只。

屋子里有一些鸡和兔子，总共有23个头，67只腿。问：有多少只鸡和多少只兔子？一步一步的推理

这是一个经典的鸡兔同笼问题，我们可以通过代数方程来解决。假设鸡的数量是 x 只，兔子的数量是 y 只。根据题意，有两个条件：

1. 总共有23个头：每只鸡和每只兔子都有一个头，因此：

$$x + y = 23$$

2. 总共有67只腿：每只鸡有2只腿，每只兔子有4只腿，因此：

$$2x + 4y = 67$$

步骤 1：解第一个方程

从第一个方程 $x + y = 23$ 中，解出 y：

$$y = 23 - x$$

步骤 2：代入第二个方程

将 $y = 23 - x$ 代入第二个方程 $2x + 4y = 67$ 中：

$$2x + 4(23 - x) = 67$$

展开并简化：

$$2x + 92 - 4x = 67$$

$$-2x + 92 = 67$$

$$-2x = 67 - 92$$

$$-2x = -25$$

$$x = 12.5$$

但是，鸡和兔子的数量应该是整数，所以这个结果看起来不合适。我们是否需要更改原始方程的解法。

生成式大型语言模型（LLM）在做什么？

文字接龙

这门课叫什么？

机器学习

“这门课叫什么？”	LLM	机
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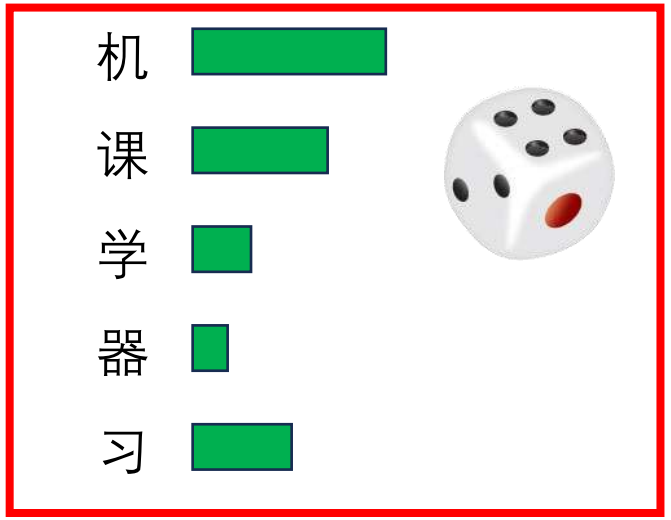
“这门课叫什么？机器学”	LLM	习
“这门课叫什么？机器学习”	LLM	END



随机性提供创造能力的同时也有不确定性

CoT给与一定确定性